ROTATION ANALYTICS INC

# Fatigue Risk Analysis: Mathematical Methodology and Validation

A Technical White Paper on the Biomathematical Fatigue Model Used by Rotation Analytics

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# 1. Executive Summary

Rotation Analytics applies a biomathematical fatigue model to rotation schedules to identify where schedules create physiological fatigue risk, even when fully compliant with collective agreements. The model is based on the Three-Process Model of alertness regulation, originally published by Torbjörn Åkerstedt and Simon Folkard in 1987 and refined through 2008.

The model produces a Fatigue Score on a 0–100 scale for every day of the rotation for every rotation line. Scores are classified into four risk levels (Low, Moderate, High, and Critical), enabling stakeholders to identify schedule segments associated with elevated fatigue exposure and make informed decisions about rotation design.

The Fatigue Score is a **relative physiological risk estimate** produced by deterministic simulation. It is intended as a decision-support tool for scheduling review, not as a diagnostic instrument, compliance determination, or safety guarantee. The model produces identical outputs for identical inputs.

This white paper documents the complete scientific foundation, mathematical methodology, engine architecture, calibration decisions, model parameters, assumptions, and limitations of the fatigue analysis delivered by Rotation Analytics.

# 2. Scientific Foundation

The Three-Process Model is the most widely cited and validated biomathematical model for shift work fatigue in the occupational health literature. Originally published by Åkerstedt and Folkard in 1987, the model has been revised and validated through 2008 across multiple occupational settings, including healthcare, transportation, and industrial operations.

The model was selected for this implementation because:

- It is published openly in peer-reviewed scientific literature; its equations and parameters are transparent and auditable
- It was developed specifically to model shift work environments
- Its parameters have been validated against real worker performance data, including healthcare and industrial settings
- It supports deterministic implementation without proprietary software
- It produces outputs that are interpretable by clients, regulators, and arbitrators

The Rotation Analytics implementation uses published equations and parameter values from the Three-Process Model literature. No novel physiological claims are introduced. The implementation's validity rests on faithful reproduction of the published model's behaviour: outputs exhibit the expected relationships between sleep pressure, circadian phase, and fatigue as documented in the source literature.

The published model describes three independent physiological processes: homeostatic sleep pressure (Process S), circadian rhythm (Process C), and sleep inertia (Process W). The Rotation

Analytics implementation activates Process S and Process C as the primary fatigue drivers, with a calibrated sleep debt modifier. Process W (sleep inertia) is documented below but is zeroed in the current implementation, as explained in the Calibration Notes.

# 3. The Three Processes

Human alertness is the product of interacting physiological systems operating simultaneously. The fatigue engine calculates each independently at every time step across the rotation.

## 3.1 Process S — Homeostatic Sleep Pressure

The longer a person is awake, the more their body builds pressure to sleep. This is a measurable neurochemical process — adenosine accumulates in the brain during wakefulness and is cleared during sleep. Process S models this mechanism as an exponential function that rises during wakefulness and falls during sleep.

During wakefulness, sleep pressure increases toward an upper asymptote of 1.0:

```
S(t) = S(t-1) + (1 - S(t-1)) × (Δt / τ_w)
```

During sleep, sleep pressure decays toward 0:

```
S(t) = S(t-1) - S(t-1) × (Δt / τ_s)
```

**Parameters:**

```
τ_w = 18 hours (wake time constant — rate of pressure buildup)
τ_s = 4 hours (sleep time constant — rate of pressure dissipation)
```

The asymmetry between these time constants is physiologically significant: pressure accumulates slowly across a shift (18-hour time constant) and clears relatively quickly during sleep (4-hour time constant). After 8 hours of sleep, pressure approaches zero. After 16 hours awake (a normal waking day), pressure has risen to a moderate level. After 24 hours of continuous wakefulness, pressure approaches the upper asymptote.

Process S is the largest single component of the fatigue score. A worker who gets shortened sleep carries residual sleep pressure into their next shift — this is the primary mechanism through which short rest periods create compounding fatigue.

## 3.2 Process C — Circadian Rhythm

The human body runs on an internal approximately 24-hour clock governed by the suprachiasmatic nucleus. This clock governs alertness independently of sleep history. Process C models this rhythm as a sinusoidal oscillator:

```
C(t) = A × sin(2π(t - ϕ) / 24)
```

**Parameters:**

```
A = 0.5 (amplitude)
ϕ = 16.8 hours (phase offset — sets peak alertness at ~16:00)
```

- **Circadian trough:** Lowest alertness occurs at approximately 03:00–05:00 hours. Night shifts spanning this window receive a circadian penalty of up to **+16 points** on the fatigue score.
- **Circadian peak:** Highest alertness occurs at approximately 16:00 (mid-to-late afternoon). Day and evening shifts receive a circadian benefit of up to –**9 points**.

Even a fully rested worker experiences a measurable reduction in alertness during the circadian trough. Circadian rhythms do not adapt rapidly to schedule changes; shifting from days to nights requires multiple days of consistent exposure, and many rotating schedules do not hold a worker on one pattern long enough for adaptation to occur.

### 3.3 Process W — Sleep Inertia (Theoretical)

The published Three-Process Model includes a third component: sleep inertia. When a person wakes from sleep, there is a short but measurable period — typically 20 to 60 minutes — during which performance is temporarily impaired. This is modelled as an exponential decay from an initial impairment value at the moment of waking:

```
W(t) = W■ × exp(-t / τ_i)
```

**Published Parameters:**

```
W■ = 0.3 (initial inertia magnitude)
τ_i = 0.5 hours (decay time constant)
```

**Implementation note:** In the Rotation Analytics implementation, Process W is set to zero (W■ = 0). The rationale for this decision is documented in Section 10 (Calibration Notes). The theoretical model is included here for completeness.

## 4. The Fatigue Score

The active processes combine into a single Fatigue Score on a 0–100 scale:

$$\text{Fatigue Score} = (S \times 100) + C + (D \times 2)$$

Where:

- **S × 100** converts the 0–1 sleep pressure value to a 0–100 base score
- **C** is the Circadian Modifier, ranging from approximately –9 (afternoon, favourable) to +16 (early morning, unfavourable)
- **D × 2** adds 2 points per hour of accumulated sleep debt

The composite score is clamped to the 0–100 range. A score of 0 represents minimal physiological fatigue. A score of 100 represents the upper bound of the model's fatigue range. All scores represent a

**representative worker**: a healthy working-age adult with no pre-existing sleep disorders, following the scheduled shift pattern without overtime or shift swaps. Scores are comparative risk indicators between schedule segments and rotation lines, not absolute predictions of individual worker performance.

# 5. Sleep Prediction Model

The model does not assume workers sleep a fixed number of hours between shifts. Instead, sleep timing and duration are determined by **rule-based deterministic logic** operating within physiological constraints derived from the Three-Process Model. Given identical schedule inputs, the sleep prediction produces identical outputs.

## 5.1 Sleep Onset

Sleep onset is triggered when two conditions are satisfied: sleep pressure exceeds a threshold value and the circadian phase is in a sleep-favourable window. During multi-day breaks, the model recognises that once sleep pressure rebuilds to a moderate level during nighttime hours (22:00–06:00), the worker will initiate sleep again, producing a normal daily sleep-wake cycle across rest periods.

## 5.2 Sleep Termination

Sleep terminates when any of the following conditions is met: sleep pressure has cleared below a threshold value, the circadian clock enters a strong alerting phase (natural morning waking), or the next shift is approaching. No fixed sleep window is assumed; sleep timing adapts to the schedule.

Night shift workers are predicted to sleep during the day because their sleep pressure is high after the night shift. Predicted sleep duration under normal conditions falls in the 6–8 hour range, consistent with published polysomnographic data for shift workers. When off-time is shortened, predicted sleep is truncated accordingly.

# 6. Sleep Debt

Process S captures within-shift fatigue accumulation. Sleep debt captures the **between-shift** effect: when consecutive rest periods are too short to fully restore the worker, each shortfall compounds into the next shift.

When predicted sleep falls below 6 hours and the gap between shifts is under 16 hours, the shortfall accumulates as sleep debt. Each hour of debt adds **2 points** to the Fatigue Score (the **k = 2** multiplier in Section 4).

Sleep debt resets to zero when the worker achieves a full sleep period of **7 hours or more**. This binary reset is a deliberate simplification: the model treats a single full sleep as sufficient recovery from accumulated short-rest debt, consistent with the model's role as a scheduling-level risk indicator rather than a clinical sleep tracker.

**Two-gate accumulation rule.** Both conditions (predicted sleep < 6 hours *and* shift gap < 16 hours) must be satisfied for debt to accumulate. A worker with several days off who sleeps less simply because they are not fatigued does not accumulate debt. The 16-hour gap threshold ensures that only genuinely compressed rest periods contribute to debt accumulation.

# 7. Simulation Methodology

The model runs a continuous simulation at 15-minute resolution across every day in the rotation, including both worked days and days off.

**Initialization.** The simulation assumes the worker begins the rotation in a fully rested state: sleep pressure is set to its post-sleep baseline (S = 0.1), sleep debt is zero, and the circadian phase corresponds to 06:00 on the first day. This assumption means the first shift block in the rotation is evaluated under favourable conditions; fatigue effects attributable to the rotation itself emerge from the second shift onward.

**Boundary conditions.** All state variables (sleep pressure, circadian phase, sleep debt) are carried forward continuously from one time step to the next. There are no resets at shift boundaries, day boundaries, or between worked and off-duty periods. The simulation terminates at the end of the last scheduled day in the rotation.

**Time Grid Specification:**

```
Resolution: Δt = 0.25 hours (15 minutes)
Example: 14-day rotation = 336 hours = 1,344 time steps
```

| Variable | Description |
|---|---|
| `Time` | Timestamp at 15-minute resolution |
| `Working` | 1 if within shift, 0 otherwise |
| `SleepOpportunity` | 1 if off-shift and in sleep-favourable window |
| `Sleeping` | 1 if predicted sleep is occurring |
| `SleepPressure` | Current Process S value (0–1) |
| `Circadian` | Current Process C value |
| `FatigueScore` | Composite fatigue metric (0–100) |

*Table 1: State variables tracked at each 15-minute time step.*

- Recovery during days off is visible in the data — you can see sleep pressure drop and fatigue scores decrease across rest periods, confirming the schedule provides adequate recovery.
- The worker's physiological state entering their next shift block is accurately carried forward from the preceding days off, rather than being artificially reset.
- On days off, the fatigue score represents the worker's resting physiological state at the end of that day — effectively, how fatigued they would be if called in at that point.

# 8. Engine Architecture

The fatigue engine follows a sequential processing pipeline:

1. **Schedule Extraction** — Read shift data from the submitted rotation file
2. **Off-Time Calculation** — Derive off-duty intervals between consecutive shifts
3. **Timeline Generation** — Build the 15-minute resolution simulation grid
4. **Sleep Prediction** — Apply sleep opportunity logic and predict sleep periods
5. **Process S Calculation** — Compute homeostatic sleep pressure at each time step
6. **Process C Calculation** — Compute circadian rhythm value at each time step
7. **Fatigue Scoring** — Combine S and C into the composite Fatigue Score
8. **Sleep Debt Modifier** — Track cumulative debt and apply upward modifier
9. **Risk Classification** — Apply calibrated risk thresholds and generate per-shift flags
10. **Output Generation** — Produce fatigue curves, risk flags, and per-shift scores

All calculations are performed in memory. Expected computation time is under one second for a standard rotation.

# 9. Risk Classification Framework

Fatigue Scores are mapped to four risk categories. These thresholds have been calibrated against real-world shift schedules to produce actionable classifications (see Section 10 for calibration rationale).

**Interpretation guidance.** Risk classifications describe the modelled physiological state of a representative worker at a given point in the rotation. They are not compliance determinations, safety guarantees, or predictions of individual worker performance. A "Low" classification does not certify a schedule as safe; a "Critical" classification does not constitute a finding of regulatory non-compliance. Classifications are intended to support scheduling review by identifying where the schedule creates conditions associated with elevated fatigue exposure.

| Score Range | Classification | Description |
|---|---|---|
| 0 – 44 | **Low** | Fatigue score is within the range expected for a standard day shift with adequate prior rest. No schedule-level concern indicated. |
| 45 – 59 | **Moderate** | Fatigue score is elevated above baseline. Commonly observed during evening shifts and consecutive day shifts where rest periods are mildly shortened. |
| 60 – 74 | **High** | Fatigue score reflects compounding sleep pressure and circadian effects. Commonly observed during night shift blocks where the circadian trough overlaps with accumulated wakefulness. |

| Score Range | Classification | Description |
|---|---|---|
| 75 – 100 | **Critical** | Fatigue score is in the upper range of the model. Produced by extended wakefulness combined with deep circadian trough and compounding sleep debt. Schedule review is indicated. |

*Table 2: Risk classification framework with calibrated thresholds.*

In addition to score-based classification, the engine flags:

- Fatigue Score exceeding High or Critical threshold during a shift
- Night shift exposure during circadian trough (03:00–05:00)
- Predicted sleep opportunity less than 6 hours between shifts
- Sleep debt accumulation across consecutive shifts

# 10. Calibration Notes

The Three-Process Model as published provides theoretical parameter values derived from laboratory studies. These parameters govern the physiological processes (Sections 3–5) and remain unchanged in the Rotation Analytics implementation. The calibration documented below addresses two separate concerns: (1) the applicability of one model component to scheduled rotations, and (2) the alignment of risk classification thresholds with the score distributions produced by real-world schedules.

## 10.1 Sleep Inertia (Process W): Excluded

The published model includes a sleep inertia component (Process W) that reduces modelled alertness for 20–60 minutes after waking. This process is relevant in on-call and emergency response environments where personnel are woken mid-sleep and must perform immediately.

Rotation Analytics analyses scheduled rotations, not on-call or interrupt-driven schedules. In a scheduled rotation, workers wake before their shift and have time to clear sleep inertia before reporting. Including Process W would reduce modelled alertness at the start of every shift for a transient physiological state that has resolved by the time the worker is on duty. This does not reflect the scheduling context the model is applied to.

Accordingly, W■ is set to 0 in the current implementation. The published equations and parameters are preserved in Section 3.3 and the Mathematical Appendix. If on-call or emergency response schedules are analysed in the future, Process W can be activated with published parameters without modifying the underlying model.

## 10.2 Risk Classification Thresholds: Adjusted

The published literature does not prescribe specific risk classification thresholds for the Three-Process Model's composite output. The thresholds commonly cited in earlier implementations (approximately 0–29 / 30–49 / 50–69 / 70–100) were derived from laboratory alertness data and do not account for the

score distributions that the model produces when applied to real-world multi-day rotation schedules.

When these lower thresholds were applied to actual healthcare rotation schedules, the majority of standard shifts classified as Moderate or High. This included schedule patterns that are well-established across multiple healthcare employers. The thresholds were not discriminating between schedule segments that differ meaningfully in fatigue exposure.

The classification thresholds were adjusted to align with the score distributions observed across a reference set of real rotation schedules:

- **Low (0–44):** Encompasses the score range produced by standard day shifts with adequate prior rest
- **Moderate (45–59):** Captures the elevation observed during evening shifts and consecutive day shifts with mildly shortened rest
- **High (60–74):** Corresponds to the compounding effects observed during night shift blocks with circadian trough exposure
- **Critical (75–100):** Reserved for the upper score range, produced only by extended wakefulness combined with deep circadian trough and compounding sleep debt

These adjusted thresholds produce classifications where each risk level corresponds to a distinct scheduling pattern. The model equations and physiological parameters are unchanged; only the interpretive boundaries applied to the composite score were adjusted.

# 11. Model Assumptions and Limitations

## 11.1 Intended Use

The fatigue analysis is a **schedule-level decision-support tool**. It is intended to identify where a rotation creates conditions associated with elevated fatigue exposure, enabling informed discussion about schedule design. It is not a diagnostic instrument, a compliance determination, a safety certification, or a substitute for clinical assessment of individual workers.

## 11.2 What the Model Can Do

- Estimate fatigue levels for a representative worker based on a given schedule
- Identify shifts and time windows associated with elevated physiological fatigue exposure
- Compare the fatigue profiles of different scheduling options on a consistent basis
- Detect cumulative fatigue accumulation across a rotation that point-in-time rules cannot identify

## 11.3 What the Model Cannot Do

- Account for individual differences in sleep need, chronotype, or health conditions
- Measure actual fatigue in a specific worker; it estimates fatigue from schedule inputs only

- Replace clinical or occupational health assessment for a specific individual
- Guarantee that a low-risk schedule will produce no fatigue-related incidents

## 11.4 Key Assumptions

- **Representative worker.** The model represents a healthy working-age adult with no pre-existing sleep disorders. Individual variation requires individual data (actigraphy, sleep diaries) that is not available in a scheduling context.
- **Schedule as provided.** The model assumes the worker follows the scheduled shift pattern as submitted. It does not account for overtime, shift swaps, or absences.
- **Relative risk indicators.** Fatigue Scores are comparative indicators between schedule segments and rotation lines. They are not absolute predictions of individual worker performance or impairment.
- **Informational only.** The analysis does not constitute medical, safety, or legal advice. It is provided as decision-support information for scheduling review.

# 12. Deliverable Format

Fatigue analysis results are delivered alongside compliance findings. The fatigue deliverable includes:

- **Per-line fatigue scores:** average, minimum, and maximum scores for each rotation line.
- **Risk classification** for each line based on average and maximum scores.
- **Rotation-wide trend visualisations** (sparklines) showing the complete fatigue and recovery cycle across every day of the rotation — both worked days and days off.
- All sparklines use the same **0–100 vertical scale** and are directly comparable across rows.
- **Risk flags** for night shift circadian trough exposure, shortened sleep opportunity, and sleep debt accumulation.

# 13. Mathematical Appendix

Complete mathematical specification of the model as implemented.

### A. Process S — Homeostatic Sleep Pressure

During wakefulness:

```
S(t) = S(t-1) + (1 - S(t-1)) × (Δt / τ_w)
```

During sleep:

```
S(t) = S(t-1) - S(t-1) × (Δt / τ_s)
```

```
τ_w = 18 hours τ_s = 4 hours Δt = 0.25 hours
```

### B. Process C — Circadian Rhythm

```
C(t) = A × sin(2π(t - ϕ) / 24)
```

```
A = 0.5 ϕ = 16.8 hours
```

### C. The Fatigue Score

```
FatigueScore(t) = (S(t) × 100) + C(t) + (SleepDebt × 2)
```

```
FatigueScore(t) = max(0, min(100, FatigueScore(t)))
```

### D. Sleep Debt

```
IF PredictedSleep < 6h AND ShiftGap < 16h:
SleepDebt += (6 - PredictedSleep)
IF PredictedSleep >= 7h:
SleepDebt = 0
```

### E. Sleep Prediction

```
Sleep Onset: Off-shift AND SleepPressure high AND
CircadianPhase in sleep-favourable window
Sleep End: SleepPressure cleared sufficiently OR
Circadian alerting phase OR
Next shift approaching
```

### F. Process W — Sleep Inertia (Zeroed)

```
W(t) = W■ × exp(-t / τ_i)
```

```
W■ = 0 (zeroed in current implementation)
Published: W■ = 0.3, τ_i = 0.5 hours
```

# 14. References


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# Rotation Analytics Inc

A division of Visser Ventures Corp.

PO Box 2234, Crossfield, Alberta, Canada

hello@rotationanalytics.ca | (403) 506-3636

rotationanalytics.ca

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